

Investigation and Evaluation of Land Use–Land Cover Change Effects on Current and Future Flood Susceptibility

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Abstract: One of the most remarkable factors affecting the flood susceptibility in watersheds is the change in land use and land cover (LULC). Moreover, machine learning (ML) methods showed reliable efficiency in predicting LULC and preparing flood susceptibility maps. Therefore, this study aims to predict LULC of the future using land change modeler (LCM), and then investigate the effects of LULC changes on the flood susceptibility map using ML algorithms in the Talar watershed, Iran. In this regard, LCM model based on artificial neural network (ANN) and Markov chain analysis used to predict LULC in the future. Then, the effect of LULC changes on flood susceptibility was investigated using some ML algorithms, including ANN, logistic regression (LR), random forest (RF), and support vector machine (SVM). These four algorithms were used for the first time simultaneously to generate flood susceptibility maps in the present and future. Nine influencing factors, including altitude, slope, distance from rivers, land use and land cover, topographic wetness index, stream power index, profile curvature, rainfall, soil type, and 200 flood and nonflood locations, were utilized to create the flood susceptibility map. The selected ML models were assessed using root mean square error (RMSE) and coefficient of determination (R^2). RMSE values showed high performance of SVM, LR, RF, and ANN models with values of 0.125, 0.175, 0.1920, and 0.2834, respectively. Furthermore, R^2 values of those models were computed as 0.9373, 0.8773, 0.8667, and 0.8525, respectively. Overall, the findings demonstrated the performance of the LCM method in forecasting the LULC map and the capability of the selected ML algorithms to produce flood susceptibility maps in the study area. In cities and watersheds all over the world that are dealing with LULC changes and a possibility of flooding, the findings and algorithms used in this study can be applied. This aids officials and decision makers to make more effective and efficient decisions in flood prevention and management. **DOI:** 10.1061/NHREFO.NHENG-1854. © 2023 American Society of Civil Engineers.

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Introduction

Among the various natural disasters, floods are considered as one of the most devastating catastrophes with huge damages (Tehrany et al. 2015). Floods are the most common natural disasters worldwide (Mojaddadi et al. 2017). Between 1960 and 2014, floods accounted for 34% of all recorded natural disasters worldwide, resulting in more than \$2.5 billion of financial losses and 1,254 deaths per year (Zeleňáková et al. 2019). Moreover, the frequency and magnitude of floods are projected to increase by 2050, which could cause huge financial losses (approximately \$1 trillion) (Rahman et al. 2021). This phenomenon usually takes place as the result of heavy rainfall, which causes rivers to overflow and temporarily overwhelm the surrounding areas (Merz et al. 2007). On the other hand, global climate change, urbanization, and changes in land use and land cover (LULC) increased the risk of floods globally by more than 40% over the past 20 years, and they may continue to do so in the future. The United States, Asia, and Europe will experience the

highest increase in the flood risk (Andaryani et al. 2021; Fahad et al. 2017, 2020; Nazari et al. 2022). Also, increased environmental degradation and LULC change brought on by population expansion, deforestation, and urbanization are the major causes of floods (Nachappa et al. 2020b). Changes in LULC may modify the discharge characteristic, runoff volume, and water quality, which can upset the hydrological equilibrium (Avand and Moradi 2021). Furthermore, the changes in LULC affect climatic systems and in turn affect recurrence and rainfall characteristics (Szwagrzyk et al. 2018). LULC will also have a direct impact on runoff from a specific rainfall (Tellman et al. 2016).

LULC is one of the time-varying inputs of flood susceptibility, which can be predicted by different models. LULC prediction using remote sensing (RS) and geographic information system (GIS), has many advantages in simulating different hydrological and environmental scenarios and phenomena in large-scale watersheds due to the ease of producing information layers and its wide application. Different studies investigate the prediction of LULC and its effects on various environmental and hydrological events. In this regard, various models were applied to predict LULC and monitor its changes. For instance, Sahoo et al. (2018) and Saha et al. (2021a), respectively, employed dynamic conversion of land use and its effects (Dyna-CLUE) to build a land-use suitability zone (LSZ) map for agricultural sustainability and to prepare a map to predict multi-hazard susceptibility such as landslides.

Sannigrahi et al. (2020), Feng et al. (2020), and Xu et al. (2021), respectively, assessed ecosystem services of a region, carbon emissions, and dynamic risk of coastal floods in the urban area by predicting LULC utilizing the cellular Automata-Markov (CA-Markov) model. Furthermore, Kafy et al. (2021) used the cellular

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automata-artificial neural network (CA-ANN) model to predict the future LULC and consequently to predict the land surface temperature (LST). Mirakhorlo and Rahimzadegan (2021) investigated soil erosion and future sedimentation using the land change modeler (LCM) and the GEO-spatial water erosion prediction project (GEOWEPPE).

The predicted LULC maps using various methods can be used in modeling, simulation, and forecasting of various hydrological and environmental phenomena such as floods. Meanwhile, the destructive effects of floods can be decreased by improved forecasting, awareness (early warning), and preparation of susceptibility maps. Predicting the location of flood-prone areas and using flood susceptibility maps is crucial to improving urban planning (Darabi et al. 2019). Authorities can estimate where flood zones will be and how severe they will be thanks to spatial modeling and flood susceptibility mapping. Therefore, utilizing a systematically designed framework and optimal models, decision-makers can develop effective flood prevention and control plans that are successful (Darabi et al. 2021). Hence, preparing a flood susceptibility map is an essential part of flood prevention and control (Avand and Moradi 2021).

In recent years, numerous methods were developed and applied to prepare flood susceptibility maps. These methods include hydrological and hydraulic models, multi criteria decision analysis (MCDA) methods, and machine learning (ML) methods. Some hydrological and hydraulic models such as Hydrologic Engineering Center–Hydrologic Modeling System (HEC-HMS) or Hydrologic Engineering Center–River Analysis System (HEC-RAS) (Alexakis et al. 2014; Zeleňáková et al. 2019; Zope et al. 2016), Two-Dimensional Flood Routing Model (FLO-2D) (Apollonio et al. 2016; Kefi et al. 2018; Mishra et al. 2018), Soil & Water Assessment Tool (SWAT) (Guo et al. 2021; Marhaento et al. 2018), and Storm Water Management Model (SWMM) (Alamdari et al. 2022), are integrated with RS and GIS for data collection and spatial analysis and rely on accurate hydrological and hydraulic data. Applying these models in places with lack of available data is challenging (Eini et al. 2020). These models also have additional constraints, including the requirement for time and precise calibration, which make it difficult to generate accurate estimations of the area (Rahman et al. 2019). Additionally, the accuracy and uncertainty of the model in these techniques will be further impacted because the simulation of hydrological processes is influenced by a variety of factors, including model parameter, structure, and input data (Ma et al. 2019).

Among MCDA methods, analytic hierarchy process (AHP) and analytical network process (ANP) were widely used in flood susceptibility mapping (Hagos et al. 2022; Mahmoud and Gan 2018). The disadvantage of these models is that in these methods, the allocation of index weight is a critical step in quantifying the risk and, therefore, is based on the experience and expertise of experts in that particular field (Chen et al. 2021). Therefore, there may be many uncertainties in the results due to the involvement of experts in the analysis (Shafapour Tehrany et al. 2019). On the other hand, ML methods were extensively used in modeling natural events and different hydrological fields, such as groundwater potential mapping (Pal et al. 2020), floods (Rahmati et al. 2019), droughts (Saha et al. 2021b), and landslides (Nachappa et al. 2020a). Among the ML algorithms used in the preparation of the flood susceptibility map, the decision tree (DT) algorithms (Tehrany et al. 2019), artificial neural network (ANN) (Rahman et al. 2019), logistic regression (LR) (Li et al. 2019), random forest (RF) (Esfandiari et al. 2020), support vector machine (SVM) (Roy et al. 2020), eXtreme gradient boosting (XGBoost) (Chen et al. 2021), and boosted regression tree (BRT) (El-Haddad et al. 2021) were widely used. Each of the ML algorithms can be used independently or together and in some cases

in combination. ML models are a subbranch of artificial intelligence; these models can train and improve themselves using historical events and prevent from mental determination of weights by discovering the relationship between flood risk and explanatory factors. Hence, there is no need for the experiences and judgments of experts to determine the importance of predictor variables. Also, ML methods can quickly find patterns and trends in huge volumes of data and therefore make better decisions with continuous improvement during training (Ma et al. 2019; Rahmati et al. 2019). In ML algorithms, at the same time that the models are trained, they are prone to overfitting, and the choice of the number epochs can affect the training process and the acquired results.

As previous studies have shown, MCDA methods have many uncertainties for flood susceptibility mapping due to their reliance on expert opinions (Nachappa et al. 2020b). At the same time, studies that used MCDA and ML methods simultaneously showed that ML methods have better predictive performance compared with MCDA (Khosravi et al. 2019; Nachappa et al. 2020b). To cope with the complexity introduced by flood susceptibility mapping models, ML methods have emerged as a crucial tool in flood susceptibility mapping (Hosseini et al. 2020). Meanwhile, it is possible to forecast future LULC modifications using LULC change models, and the future flood susceptibility maps may be created using the expected LULC. Limited studies have been conducted comprehensively on LULC prediction and the use of ML models to develop future flood susceptibility maps. Furthermore, very limited studies have predicted the future LULC map by the LCM model, in which the effect of LULC change on floods was investigated. Additionally, no research has been done to date that concurrently uses the four well-known and practical ML algorithms, including ANN, LR, RF, and SVM to assess flood susceptibility. Accordingly, the purpose of this study is to predict LULC in the future using an LCM model by ANN and Markov chain, and then analyze the effects of LULC changes on the flood susceptibility map using ML algorithms, including ANN, LR, RF, and SVM.

Material and Methods

Study Area

This study was conducted in the Talar watershed in northern Iran, where floods have caused much damage in recent years. The Talar watershed is located between 52° 35' 48.25" and 53° 23' 21.72" eastern longitudes, 35° 44' 8.59" and 36° 19' 25.2" northern latitudes. The Talar watershed is located between the three provinces of Mazandaran, Tehran, and Semnan, most of which is in Mazandaran province, Iran (Fig. 1). The watershed has a total area of 2,100.89 km² and a perimeter of 279.99 km. The Talar River, which is 100 km long, and the Kassilian River, which is 50 km long, are the two major rivers in the watershed. The altitude of the region ranges from 215 to 3,944 m above mean sea level. The maximum slope in the watershed is 75.19° and the average slope is 21.7°. This watershed has eight rain gage stations which are shown in Fig. 1, and the average annual rainfall in the whole watershed is 521 mm.

Used Data

Flood Inventory Map

To assess the risk and prepare a flood susceptibility map in an area, it is essential to analyze past flood events. Accordingly, a flood inventory map is prepared, showing the location of historical flooded areas in the watershed area, and the relationship between historical

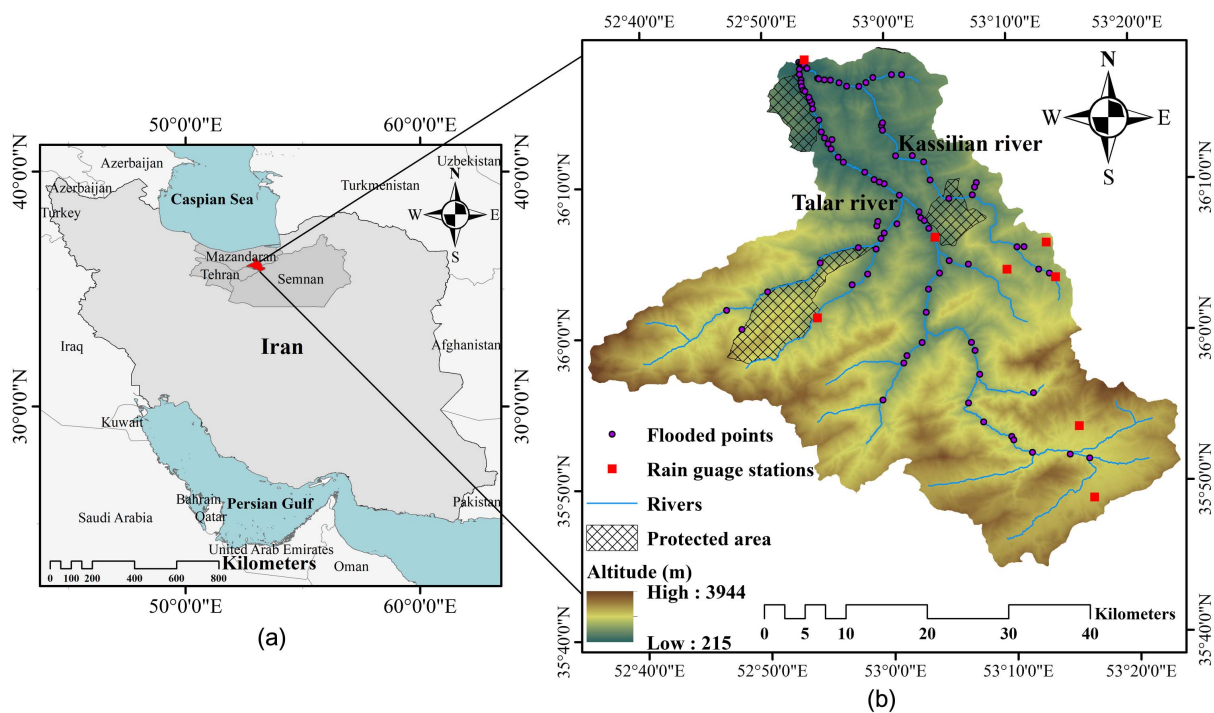


Fig. 1. The map of (a) the location of the study area; and (b) digital elevation model, protected areas, location of rain gauge stations, the location of the historical flood, and drainage network in Talar watershed.

floods and the factors that cause it. The accuracy of this map and the location of historically flooded points will have a significant influence on the flood susceptibility map (Tehrany and Kumar 2018). In this regard, 100 historical flood points were acquired for this research from a variety of sources, including field surveys, Google Earth images, historical flood records from government organizations, and subject matter experts. Besides, 100 nonflood points were prepared in the watershed, which were considered in places with low probability of floods, such as mountain peaks or areas very far from waterways. Among the flood inventory map data, 70% of flood and nonflood points (70 flood points and 70 nonflood points) were used for model training and the remaining 30% of points (30 flood points and 30 nonflood points) were utilized for testing and evaluating model performance. In this way, these two parts of the data were used to prepare and predict flood susceptibility maps using ML.

Satellite Data and Information Layers Used

To develop LULC maps of the study area, three Landsat satellite images in 2000, 2010, and 2020 with resolution of 30 m were prepared from the website of United States Geological Survey (USGS 2022), the specifications of which have been given in Table 1. To achieve more accurate LULC maps and due to the different coverage of some lands such as pasture within a year, it has been attempted to select the images in the desired years as far as possible in a similar time period. Moreover, the digital elevation model (DEM) from the

shuttle radar topography mission (SRTM) with resolution of 30 m was prepared from the website of United States Geological Survey (USGS 2022), to generate topographic information of the region. Furthermore, a map of the river basin was prepared from the Mazandaran Regional Water Company. To consider the protected areas in the LULC forecast, the map of these areas in this watershed was prepared by the Environmental Protection Organization. Also, soil type data were obtained from the General Department of Natural Resources and Watershed Management of Mazandaran Province. In addition, rainfall data for the relevant stations were obtained from the Iran Water Resources Management Company, which includes daily rainfall at eight stations within the watershed from 2005 to 2015 [Fig. 1(b)].

Methodology

The general flowchart of this study has been shown in Fig. 2. Moreover, the details of each step of the flowchart are described in the following.

Producing Influencing Factors on Flood

In general, the selection of appropriate influencing factors is a necessary and important part of the development and analysis of flood susceptibility mapping studies using ML methods. In this study, different ML methods were used in both main parts, namely LULC forecasting and flood susceptibility mapping. There are no general guidelines for selecting the factors influencing LULC and flood susceptibility. Then, in this study, 10 factors, including altitude, slope, distance from rivers, distance from forest, distance from built-up, distance from agriculture, distance from pasture, distance from bare land, distance from roads, and evidence likelihood were selected for forecasting LULC. In addition, nine factors, including altitude, slope, distance from rivers, LULC, topographic wetness index (TWI), stream power index (SPI), profile curvature, rainfall, and soil type were utilized for producing a flood susceptibility map.

Table 1. Characteristics of Landsat images of the years 2000, 2010, and 2020 used in this study

Landsat series	Sensor	Acquisition date	Path/row	Spatial resolution (m)
Landsat 5	TM	June 17, 2000	163/035	30
Landsat 5	TM	May 28, 2010	163/035	30
Landsat 8	OLI/TIRS	June 24, 2020	163/035	30, 15

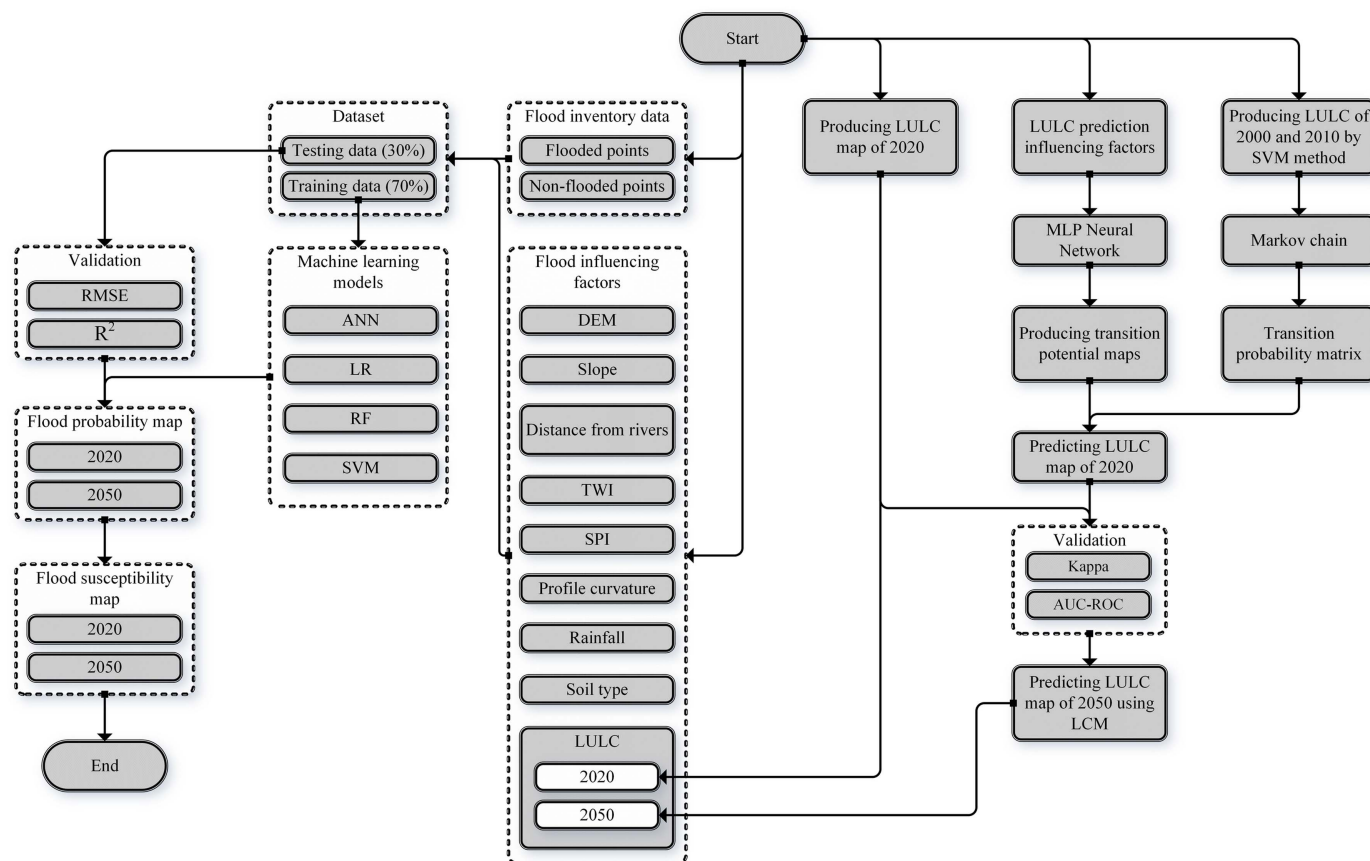


Fig. 2. Flowchart of this research.

All of these factors have been selected based on data availability, expert views, and previous studies (Arabameri et al. 2020; Avand and Moradi 2021; Nachappa et al. 2020a; Zhao et al. 2019). However, the inclusion of additional factors in modeling does not necessarily guarantee better results (Tehrany et al. 2019). Each of the used influencing information layers are introduced and described as follows.

Altitude. One of the key factors influencing flood susceptibility is altitude (Dodangeh et al. 2020). This factor is also an important factor in predicting the LULC map. Elevation and floods are generally inversely related, so that the higher the altitude, the less likely it is to occur flood in that area, and it is almost impossible to encounter flooding in very high altitude areas (Choubin et al. 2019).

Slope. The angle between height and horizontal distance is known as the slope. Maps of flood susceptibility and LULC forecasts both heavily depend on the slope of the area. An area's slope regulates and affects flood susceptibility and LULC forecasts. Increasing the slope rises the flow velocity and causes waterlogging and flooding in areas with a low and flat slope. Therefore, flat and low-slope locations are more vulnerable to flood than other places (Arora et al. 2021; Bui et al. 2019). The slope map of the area was produced from the DEM in GIS environment.

Distance from Rivers. Floods usually occur frequently along the river, and the areas and infrastructures around the river are the most prone areas for floods. Therefore, one of the most remarkable influencing factors in floods occurrence is the distance from rivers. Moreover, this parameter can be an important and effective factor in LULC forecasting that should be investigated. The distance from the river was generated using the Euclidean distance in the GIS environment.

LULC. It is clear that LULC in the past is a very important factor in predicting LULC in the future (Feng et al. 2020; Mirakhorlo and Rahimzadegan 2021; Sannigrahi et al. 2020; Xu et al. 2021). On the other hand, LULC plays a crucial role in flood susceptibility and directly and indirectly affects some hydrological processes such as permeability, evapotranspiration, and runoff (Yariyan et al. 2020). In areas with more vegetation, runoff speed is reduced and infiltration is increased, thus the flood susceptibility is reduced. The flood susceptibility will rise in bare lands and urban areas with increased impermeability levels because the infiltration rate declines and the flow velocity increases. Landsat images in the three years 2000, 2010, and 2020 were prepared to produce LULC maps. Initially, preprocessing such as radiometric and geometric corrections were performed, and then images were classified in a supervised manner using the support vector machine (SVM) algorithm to produce the LULC map for selected years. The SVM method for image classification was used by many studies and its accuracy has been confirmed compared with other image classification methods (Ibrahim Mahmoud et al. 2016; Varamesh et al. 2017). The LULC map of 2000 and 2010 was used as an input in the prediction of LULC map of 2020. The classified LULC map of 2020 is one of the influential factors for the preparation of flood susceptibility map in 2020.

Distance from LULC Classes. This factor is one of the LULC prediction model's inputs, which include distance from LULC classes, including forest, pasture, agricultural land, bare land, and built-up areas. Each of these factors can explain the changes between two base years, 2000 and 2010 that need to be investigated. All these factors were generated using GIS.

Distance from Roads. Another factor that can be effective in forecasting LULC and entered as an input to the LULC prediction model

is the distance from roads. It makes sense that between the two base years, some LULC changes would be close to the roads due to accessibility. Therefore, the distance from the roads can be an influential factor. This map was also prepared in GIS environment.

Evidence Likelihood. Evidence likelihood is an experimental and qualitative probability of change in LULC classes between the previous and next map and is used to convert classified variables, such as conversion from one LULC class to another, with numerical values (Leta et al. 2021). This factor is created by determining the relative frequency with which the various LULC categories are transferred from one year to the other. According to this, changed pixels are less likely to change in the future.

Protected Areas. Protected areas are among the factors that can be influential in forecasting LULC in the future. Using a Boolean function, a very small value (very close to zero) is assigned to the protected areas and a value of one is assigned to the other areas where there is a possibility of a future change of LULC to create a map of protected areas.

Topographic Wetness Index (TWI). This index is considered as an important topographic factor that quantifies topographic control over hydrological processes, including floods (Khosravi et al. 2018). TWI shows the flow accumulation at any point in the watershed and the trend of water falling by gravity (Gokceoglu et al. 2005). TWI values are usually higher in flooded areas and are calculated using the following equation (Beven and Kirkby 1979; Moore et al. 1991):

$$TWI = \ln \left(\frac{A_s}{\tan \beta} \right) \quad (1)$$

where A_s represents the specific area of the watershed in m^2/m in the watershed; and $\tan \beta$ is the slope angle in degree at that point. This parameter was produced using DEM data.

Stream Power Index (SPI). Another topographic characteristic used in this study is the SPI, which is used as an indicator of the erosive power of water flow at a given point in the topography (Gokceoglu et al. 2005). The following equation is used to calculate this index:

$$SPI = A_s \times \tan \beta \quad (2)$$

Profile Curvature. This factor is the curvature parallel to the slope, which shows the type of land shape in that area, and has been used to investigate flood susceptibility (Roy et al. 2020). DEM data were also used to derive this component. A value of zero implies that the surface is flat at that location and has no curvature, whereas positive and negative values denote concavity and convexity, respectively.

Rainfall. It is clear that rainfall is one of the most important factors in flood susceptibility and the more rainfall in an area, the more likely it is to flood in that area, assuming that other conditions are the same. After calculating the average annual rainfall of eight rain gauge stations within the watershed for 10 years, interpolation was performed by the kriging method. Many studies, including Chen et al. (2020), have used the annual average to include rainfall data into the study.

Soil Type. Another factor that can affect the flood susceptibility is the type of soil in the area and its permeability. At one point, more soil permeability reduces the chance of runoff and subsequent floods, and vice versa. The soil type factor of the region, which indicates its permeability in any portion of the watershed in cm/s , was produced after compiling the soil characteristics of the watershed.

Modeling of LULC Changes

Eastman et al. (2005) initially developed the LCM. In this research, LCM was used to model the changes and predict the LULC map. In this study, multi-layer perceptron (MLP) and Markov chain were

used to perform LCM. The overall process of LCM includes three stages of change analysis, transition potential modeling, and change prediction (Eastman 2015).

Several maps are required as an influential factor for predicting LULC by MLP in LCM. To determine the importance of influencing factors used as MLP input, Cramer's V measure was used, which measures the correlation between two variable classes. This numerical criterion has a value between 0 and 1, with a larger value indicating a stronger correlation and higher importance of the relevant factor. In general, factors with Cramer's V higher than 0.15 are useful, and if this value is higher than 0.4, they will be good for selection as MLP input (Eastman 2015, 2016). It should be noted that this measure does not guarantee the strong performance of the variables, and in fact a variable may be less than 0.15, but it is expertly recognized that it is better to enter the model as an input. But, in general, this measure can be used to determine which variables enter the MLP model as an influencing factor (Leta et al. 2021). After using this measure, nine variables with the highest value of Cramer's V, including altitude, slope, evidence likelihood, distance from roads, and all distance layers from LULC classes were included in the MLP model as influential factors.

MLP was used to generate transition potential maps, which has suitable performance for studying LULC change (Eastman 2015). Five submodels in MLP neural network were used to model the transition potential and generate its maps. These five submodels are forest to pasture, forest to bare land, pasture to forest, pasture to bare land, and bare land to pasture. In LCM, the MLP neural network is a feedforward neural network with one input layer and one output layer, and a number of neurons in the hidden layer, that in this study, the number of them was obtained using trial and error. The output of this section, along with the output of the Markov chain, was used as input to the forecast section. The Markov chain was used to generate the transition probability matrix. The LULC maps of 2000 and 2010 were used to calculate the transition probability matrix by the Markov chain. The Markov chain is a process in which the state of a system can be predicted by knowing its previous state, and the probability of changes from one state to another can be determined (Alexakis et al. 2014; Eastman 2015). The probability of transition of LULC change in the matrix of the Markov chain indicates the probability of transfer from one LULC class to another in two different time periods.

The area of different LULC classes for 2020 was estimated by combining the Markov transition probability matrix and the transition potential map derived from the MLP model and considering protected areas. Then, to establish if the LCM was appropriate for the chosen scenario to forecast the future LULC map, the accuracy of the created map of 2020 using the LCM was assessed with the LULC map of 2020 classified using the SVM technique as ground truth data. After achieving satisfactory results, the LCM model with the scenario mentioned above and the reference LULC maps of 2010 and 2020 were used to forecast the LULC map in 2050.

Flood Susceptibility Modeling

In this study, several ML algorithms, including ANN, LR, RF, and SVM have been used to model floods and prepare flood susceptibility maps. All of these algorithms are implemented in the Python 3.8.3 programming environment. Flood influencing factors in these algorithms include altitude, slope, distance from rivers, LULC, TWI, SPI, profile curvature, rainfall, and soil type, which were used as the inputs along with the flood inventory map. In fact, each of the influencing factors were entered into the ML algorithms as features. The used ML algorithms were implemented as a classification algorithm in supervised learning. In this case of ML, the targets have labels, which in this study are flood and nonflood points. In each of these

algorithms, two flood susceptibility maps are generated as the output. The real LULC map of 2020, which was classified using the SVM technique, and other influencing factors, were used to create the first map, which is the flood susceptibility map for 2020. The second map, called the 2050 flood susceptibility map, was created by utilizing the 2050 LULC map that the LCM model generated and other influencing factors. The step-by-step process for the proposed method is shown in the flowchart illustrated in Fig. 2. Because the nine influencing factors used in this study change in terms of value in different ranges, data normalization was done so that the data are in the same range and better results are obtained. There are several methods for data normalization. In this study, the following method was used, which is one of the most common methods for data normalization and in which the values will be between 0 and 1:

$$Z_i = \frac{X_i - X_{\min}}{X_{\max} - X_{\min}} \quad (3)$$

where Z_i represents the normalized value of the desired parameter; X_i denotes the initial value of the parameter before normalization; and $X_{\min}$ and $X_{\max}$ show the minimum and maximum values of the parameter, respectively. The algorithms used to prepare the flood probability map are described below. After running each of the algorithms, a flood susceptibility map with values between 0 and 1 was obtained, the values of which indicate the probability of flood occurrence in the desired pixel. Then, using the natural breaks (Jenks) method, it was classified into five classes, including very low, low, medium, high, and very high risk, and a flood susceptibility map was obtained.

Artificial Neural Network

The algorithms known as artificial neural networks (ANNs) are made up of a black box and a mathematical model that are inspired by the structure and operation of the neural networks in the human body. ANN is one of the most popular ML methods used to solve various types of complex problems (Pal et al. 2020). In this study, the used general structure of back propagation (BP) ANN algorithm, consists of three main parts: input layer, hidden layer, and output layer. The input layer receives data from various sources. Each of the neurons in the input layer will be assigned to one of the influential factors of the flood. Therefore, the number of input layer neurons will be equal to nine, and the number of output layer neurons was considered as one. The values of the input layer are transferred by nodes to the next layer in the feedforward path. The connection between the input and hidden layers and the hidden and output layers is established through weights. After applying weights to each of the neurons and summing the values obtained, the activation function is used to obtain the output of the corresponding neuron. The number of hidden layers and the number of their neurons, the activation function of the hidden layer neurons, and the activation function of the output layer neurons were obtained through the trial and error method.

Logistic Regression

LR is a technique derived from the field of statistics by ML. This method is a nonlinear regression model for solving binary problems (problems with classes of two values) developed by Cox (1958), and was first introduced by McFadden (1973) to measure the probability of occurrence of any catastrophe in an area using a special formula to generate influential factors. LR assesses a dependent variable of an event, such a flood, and determines the relationship between the dependent variable (flood) and the independent variables (flood influencing factors). Through LR analysis, a regression

relationship is established between flood inventory and flood influencing factors (Shafapour Tehrani et al. 2019). Flood inventory is a binary factor that indicates the presence or absence of flood. The probability of a flood occurring (P) has values between 0 and 1, and the closer it is to 1, the higher the probability of a flood occurring. The main relationship of the logistic regression model is as follows:

$$P = \frac{1}{1 + e^{-z}} \quad (4)$$

$$z = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n \quad (5)$$

where P = probability of flood occurrence; β_i ($i = 0, 1, 2, \dots, n$) represents the model regression coefficient; and x_i ($i = 0, 1, 2, \dots, n$) indicates the influencing factors of the flood.

Random Forest

RF was first proposed by Ho (1995) and developed by Breiman (2001) using the random subspace method. RF is a ML technique in which input datasets are classified based on multiple decision tree sets and trained from multiple decision trees to increase model accuracy. This algorithm was created as a potent collection of ML algorithms that combines regression, classification methods, and statistical theory (Darabi et al. 2021). Because of its quick processing speed and superior classification results, RF has attracted a lot of interest lately (Nachappa et al. 2020b). In RF, each decision tree is taught using a random sample selected from the training data set (Breiman 2001). Decision trees in RF use controlled variance, which improves the quality of the prediction and solves the problems of overfitting (Ho 1998). Classification trees provide the power to select and accurately classify the vote of the majority of trees throughout the forest. To run the RF model, one of the most important parameters to be selected by the user is the number of trees, which is usually 500 to 1,000 trees (Avand and Moradi 2021). The number of forest trees was obtained using trial and error method in this research.

Support Vector Machine

SVM is one of the most powerful supervised ML algorithms for classification and regression that can deal with problems such as nonlinearity and high dimensions (Hong et al. 2020). SVM is also known as the maximum margin method which can provide better performance and results with limited number of data (Nachappa et al. 2020b). This algorithm was developed based on statistical learning theory in which input data is transferred to a new feature space, and a decision function is created in this space through the optimal hyperplane (Nachappa et al. 2020a). Creating a hyperplane to separate the input data and then selecting the best hyperplane to form the maximum distance and create the maximum possible margin is the basis of SVM. The hyperplane will be in the center of the maximum margin obtained. In this algorithm, to create an optimal hyperplane, the kernel function is used, which takes each of the data as input and adds a dimension or feature to them, thus making it easier to form the maximum distance and identify the optimal hyperplane. The decision function is used to classify new data as:

$$g(x) = \text{sign} \left(\sum_{i=1}^n y_i \alpha_i K(x_i, x_j) + b \right) \quad (6)$$

where α_i = Lagrangian coefficient; and $K(x_i, x_j)$ = kernel function. The performance of the SVM model is affected by the kernel function and will have a direct effect on the accuracy of the results. The most important kernel functions used include linear kernel,

Sigmoid kernel, polynomial kernel, and radial basis function (RBF) kernel. In this study, to achieve the best result and the highest accuracy and prepare a better flood susceptibility map, all four types of kernels mentioned were investigated and the best kernel was used.

Model Evaluation Methods

Evaluation Methods of LCM Model

Demonstrating goodness of fit and evaluation using appropriate accuracy criteria is important in a spatial forecasting model. There is no best criterion, to evaluate the performance of the LULC forecasting model. However, in this study, the kappa coefficient and area under curve—receiver operation characteristic (AUC-ROC) have been used by comparing the map produced in 2020 by LCM with the real map of 2020 classified by SVM, that has shown its efficiency in similar studies (Avand and Moradi 2021; Mirakhorlo and Rahimzadegan 2021; Xu et al. 2021; Zare et al. 2017). The kappa coefficient is calculated from the following equation:

$$Kappa = \frac{p_{obs} - p_{exp}}{1 - p_{exp}} \quad (7)$$

where p_{obs} represents the accuracy of the predicted map compared with the real map; and p_{exp} is the possibility of a random agreement. The kappa value is between 1 and -1 , the ideal value of which is 1, indicating the complete similarity of the two images. A value of zero indicates a random prediction and a value of -1 indicates a high prediction error. According to Landis and Koch (1977), Kappa values of 0–0.2, 0.2–0.4, 0.4–0.6, 0.6–0.8, and 0.8–1 describe the level of agreement between model and reality as poor, fair, average, good, and almost excellent, respectively.

To improve the accuracy of the assessment, the AUC of the ROC curve was also employed. Plotting the true positive rate versus the false positive rate at various thresholds results in the ROC curve (Kim and Newman 2019). The AUC represents the overall fit, which varies from 0 to 1, where 0.5 means chance performance and 1 means perfect fit. AUC values of 0.5–0.6, 0.6–0.7, 0.7–0.8, 0.8–0.9, and 0.9–1 indicate poor, average, good, very good, and excellent performance of the model, respectively (Darabi et al. 2019). In general, AUC of ROC curve is one of the most widely used evaluation criteria to evaluate the performance of different models in different fields, which has also been used in this study.

Methods of Evaluating ML Models

Evaluating the results of the used models shows their ability to predict and their performance, which is an important step. In this study, to evaluate the results and performance of the ML models used to prepare the flood susceptibility map, the RMSE and R^2 criteria were used, which are frequently used in similar studies, especially in the field of hydrological phenomena studies (Hong et al. 2020; Khosravi et al. 2019; Liu et al. 2020; Rahman et al. 2021; Saha et al. 2021b). RMSE and R^2 are obtained by comparing and using the observed values and the actual values by the following equations:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (a_i - p_i)^2} \quad (8)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (a_i - p_i)^2}{\sum_{i=1}^n (a_i - \bar{a})^2} \quad (9)$$

where n is considered as the number of observations; a is used as the actual value; p is the value predicted by the model; and $\bar{a}$ represents the mean value of the actual values. The closer the values predicted

by the model to the actual values, leads to the lower the RMSE and the closer values to zero. Also, the value of R^2 is close to the number one indicate the ideal performance of the model used. Assessment of ML models is performed by the two criteria mentioned above in Python 3.8.3.

Results and Analysis

Prediction of LULC Map

At the first stage, the maps related to the influencing factors in prediction the LULC map as well as the flood susceptibility were produced according to Fig. 3. Then, three LULC maps of 2000, 2010, and 2020 were used to implement LCM and predict LULC map of 2020 and 2050. Using SVM method, these three maps were classified into five classes, including forest, built-up, agriculture, pasture, and bare land areas (Fig. 4). Cramer's V measure was used to select the input layers to the MLP model. Table 2 displays the values of this measure for various layers. Due to the fact that the value of Cramer's V measure was the lowest for the distance layer from rivers compared to other layers, it was not included in the MLP model in LULC prediction. Thus, altitude, slope, distance from forest, distance from pasture, distance from agriculture, distance from bare land, distance from built-up areas, distance from roads, and evidence likelihood along with protected areas were used in LULC prediction.

Using the LCM model and the LULC map of 2000 and 2010 along with the influential layers and the protected areas map, the LULC map of 2020 was predicted to include five LULC similar to the reference LULC maps [Fig. 4(d)]. The results obtained from this model show the kappa statistic of 0.82 and the AUC of 0.94 for the evaluation of the predicted LULC map for 2020, which is in the excellent range based on both criteria. In addition, the LULC map for 2050 was predicted using LCM model and the mentioned scenario to produce the LULC map for 2020 [Fig. 4(e)].

According to the predicted LULC map for 2050 using LCM, during the 30 years after 2020, the area of forest coverage in this watershed will decline by 15.09 km² and the area of built-up areas will increase by 11.44 km² (Table 3). Moreover, agricultural land use will increase by 1.21 km² compared to 2020 in 2050. Pasture and bare land uses compared to 2020 will increase and decrease 45.46 and 42.12 km², respectively.

If LULC changes over the next 30 years are similar to the changes over the past 20 years, the LULC change map between 2020 and 2050 will be as shown in Fig. 5, according to the LULC map predicted by LCM in 2050. The area of LULC change zones over 30 years can also be seen in Table 4. Changes of less than 0.05 km² have been omitted in Table 4 and Fig. 5, as these changes may be due to some modeling errors. In Fig. 5, the more destructive the LULC changes are for the occurrence of flood, the more blue color intensity the display of that type of change is. The three most destructive changes are LULC change from forest to built-up areas, forest to bare land, and forest to pasture, respectively. The conversion of a 48.15 km² area from forest to pasture is the most detrimental shift. Although the conversion of pasture to bare land is also destructive and takes place over a sizable area of 47.344 km², the impact of vegetation types on increasing water infiltration and lowering flood flow speed means that conversion of pasture to bare land will have a much smaller effect on increasing flood incidence than conversion of forest to pasture. As can be seen in Fig. 5, the change of forest land use to pasture has affected the northern part of the watershed.

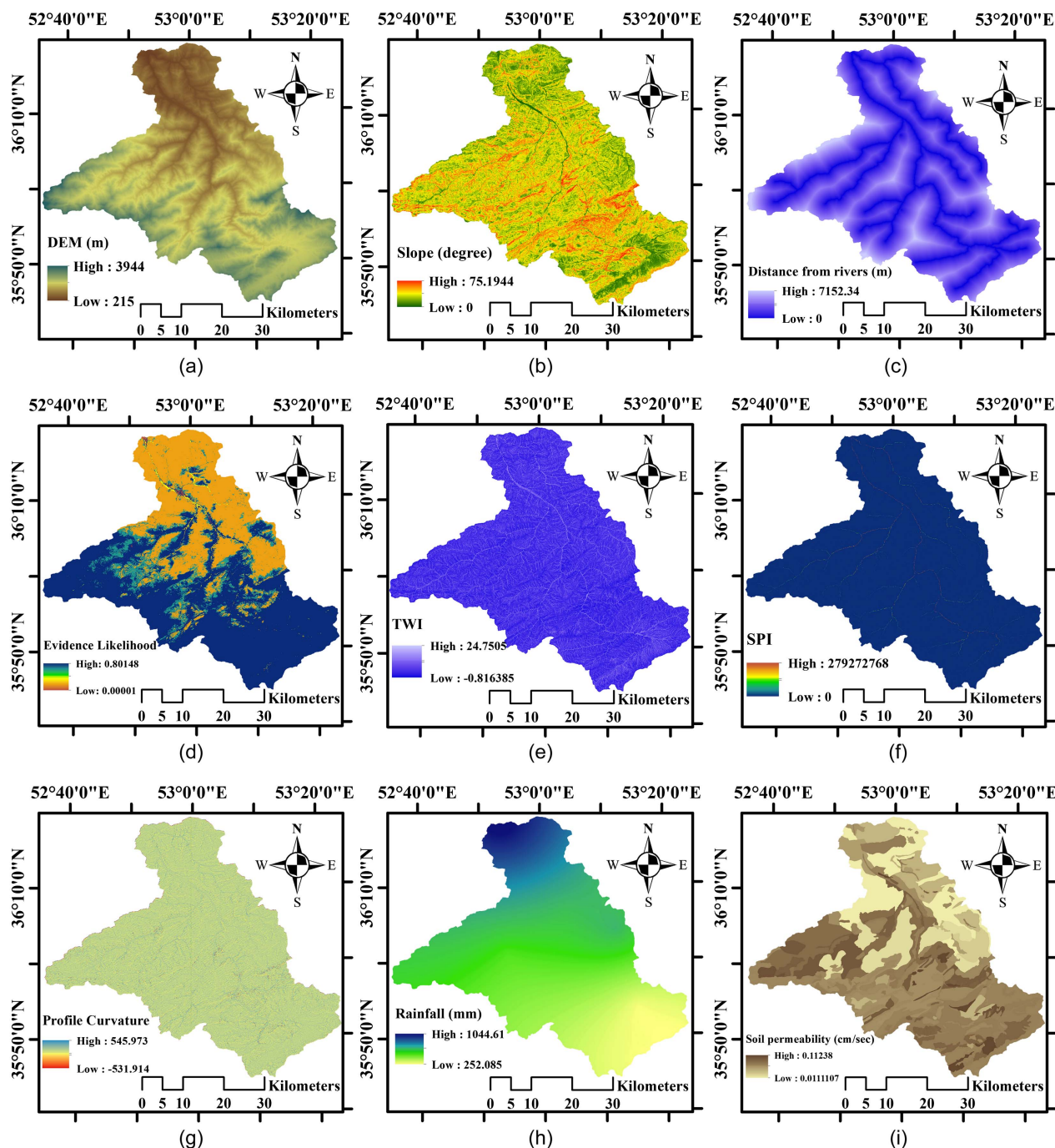


Fig. 3. The maps of (a) elevation; (b) slope; (c) distance from rivers; (d) evidence likelihood; (e) TWI; (f) SPI; (g) profile curvature; (h) rainfall; and (i) soil type.

Some LULC changes such as change from bare land to pasture as well as change from pasture to forest will also occur in the watershed. These changes can be useful changes to reduce flooding and cover 33.992 and 27.016 km² of watershed area, respectively (Table 4). These changes have been shown in reddish colors in Fig. 5. Most of these LULC changes have occurred in the central parts of the watershed and its middle belt. In general, considering the fact that the changes in the LULCs of bare land to pasture and pasture to forest to reduce the incidence of floods, 61,008 km² of the watershed, will have changes to reduce the likelihood of flood. Other changes to increase the likelihood of floods, 103.668 km² of the watershed area will have changes to increase the probability of flood occurrence.

Preparation of Flood Susceptibility Map

Analysis of Results of ML Algorithms

Four algorithms ANN, LR, RF, and SVM were used to produce the flood susceptibility map by considering nine influencing factors including altitude, slope, distance from rivers, LULC, TWI, SPI, profile curvature, rainfall, and soil type (Figs. 3 and 4). After running ANN, the best fit of this method was determined by trial and error; it consists of a hidden layer with eight neurons, a ‘tanh’ activation function for hidden layer neurons, and a sigmoid activation function for output layer neurons. Moreover, LR, RF, and SVM algorithms were implemented, that in RF algorithm, 500 decision

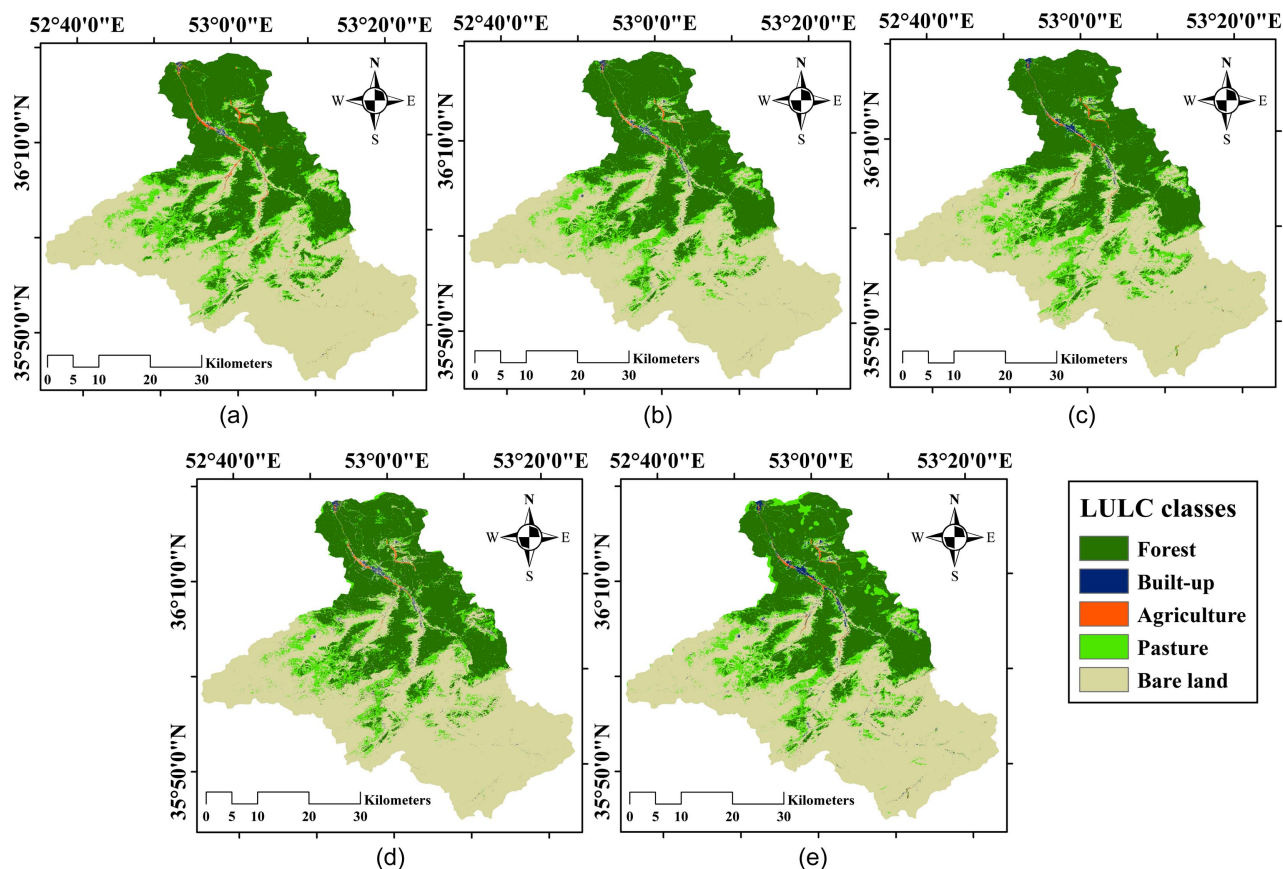


Fig. 4. LULC maps (a) classified for 2000; (b) classified for 2010; (c) classified for 2020; (d) predicted by LCM in 2020; and (e) predicted by LCM in 2050.

Table 2. Cramer's V statistic values

Influencing factor	Cramer's V
Elevation	0.3365
Slope	0.0787
Distance from rivers	0.0336
Distance from forest	0.3094
Distance from pasture	0.1938
Distance from agriculture	0.1531
Distance from bare land	0.2989
Distance from built-up	0.1146
Distance from roads	0.1231
Evidence likelihood	0.5594

trees provided the best result, and in SVM algorithm, the best performance after trial and error was obtained by RBF kernel. After implementing each of the mentioned algorithms, using the real LULC map of 2020 and other factors influencing the flood, the values of flood occurrence probability between zero and one were obtained for all pixels in the watershed. The closer this value is to one, the more likely a flood will occur at that point. Using LULC map predicted in 2050 by the LCM model, and other factors influencing the flood, similarly, the flood probability map in 2050 was obtained by ANN, LR, RF, and SVM algorithms with the mentioned specifications for each one of the algorithms mentioned above.

Then, a classification map of flood susceptibility was prepared in the studied watershed for a better interpretation of flood prone areas (Fig. 6). The Natural Breaks (Jenks) method was utilized in this research, in which each of the classes have the most differences

Table 3. The difference in the area of LULC classes between 2020 and predicted for 2050

Row	LULC classes	LULC area in 2020 (km ²)	LULC area in 2050 (km ²)	Difference area (km ²)	Percentage change (%)
1	Forest	658.33	643.24	-15.09	-2.35
2	Built-up	11.44	21.97	+10.53	+47.93
3	Agriculture	4.54	5.75	+1.21	+21.04
4	Pasture	208.31	253.77	+45.46	+17.91
5	Bare land	1218.3	1,176.18	-42.12	-3.58

from each other, and the values that are placed in the same class will have the most similarities to each other. Finally, a flood susceptibility map was prepared for each of the ML algorithms in 2020 and 2050, which includes five flood classes with very high, high, medium, low, and very low probability of occurrence (Fig. 6).

As shown in Fig. 6, according to all algorithms, flood susceptibility in the northern regions is higher than the central and southern regions. Flood susceptibility maps for both 2020 and 2050 are very similar for all algorithms. But, as it seems qualitatively, the factor of altitude in the watershed is one of the most influential factors in ANN algorithm, and the factor of distance from rivers seems to be much more effective in other algorithms.

Evaluation of ML Models

The evaluation process was performed by comparing the observed flood data and the values of the flood susceptibility map at the desired point. The predictions of the models used were evaluated by

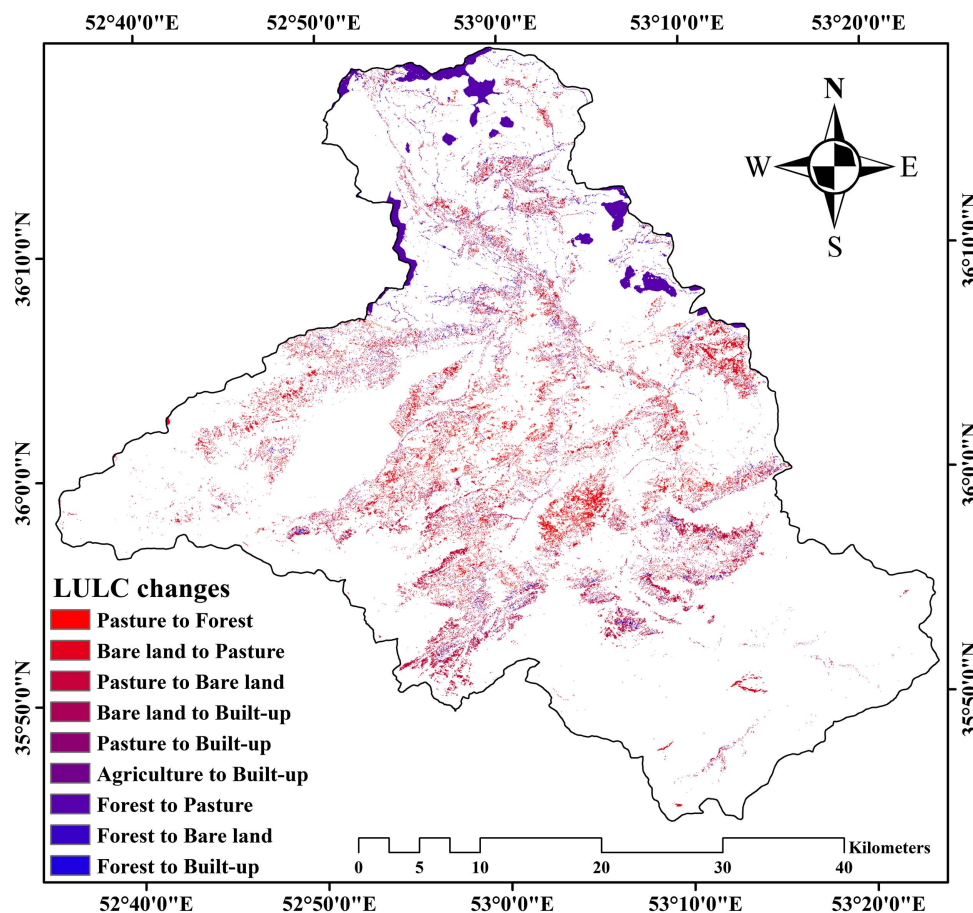


Fig. 5. Map of changing different LULCs between 2020 and 2050.

Table 4. Changed LULCs from 2020 to 2050

Row	LULC changes	Area (km ²)
1	Pasture to forest	27.016
2	Bare land to pasture	33.992
3	Pasture to bare land	47.344
4	Bare land to built-up	6.469
5	Pasture to built-up	1.084
6	Agriculture to built-up	0.224
7	Forest to pasture	48.153
8	Forest to bare land	0.291
9	Forest to built-up	0.103

two criteria of RMSE and R^2 (Table 5). Based on the RMSE criterion, the SVM model showed the best performance with an accuracy of 0.1252. Then, the LR algorithm with a value of 0.1752 and the RF and ANN models with a value of 0.1920 and 0.2834 will follow them. Moreover, using the R^2 criterion, the SVM algorithm with the accuracy of 0.9373 has the best result and then the LR, ANN and RF algorithms with the values of 0.8773, 0.8667, and 0.8525, respectively, showed good performance. Although all the models used in this research provided good accuracy, similar to other studies that have been done using different ML algorithms, the SVM model in the watersheds where the number of data is limited, provided better accuracy.

Evaluation of the Effects of LULC Changes

After preparing the flood susceptibility map (Fig. 6), the difference in area of each flood susceptibility class between 2020 and 2050 was calculated (Fig. 7). As observed in Fig. 7, except for the ANN

method, in which very low risk areas will increase, in all algorithms over the next 30 years, the percentage of areas with very low flood susceptibility will decrease and the areas with very high flood susceptibility will increase.

The results and predictions of the models used show that from years 2020 to 2050, areas with low and very low flood susceptibility will decrease and areas with high and very high flood susceptibility will increase. As a result, considering that only LULC changes were taken into account for prediction of the 2050 flood susceptibility map and all other influencing factors remained constant, it was found that the negative effects of land use changes over the ensuing 30 years, particularly the destruction of forest areas and vegetation and an increase in built-up areas and bare lands, would increase the likelihood of flooding and expand the areas with high and very high flood risk.

Considering that among the four ML algorithms used, the SVM model prediction has the highest accuracy, the map of flood susceptibility changes in the watershed from 2020 to 2050 was prepared based on this model (Fig. 8). As can be seen in this figure, the higher the intensity of the change from an area with lower flood susceptibility to an area with higher flood susceptibility, the more blue color intensity the display color will be, and vice versa.

Three of the most destructive changes are areas with moderate to very high, low to high, and very low to medium flood susceptibility, respectively, which cover 110,700, 150,300, and 2,700 m² of the watershed area, respectively. Also, destructive changes with less effect, including areas with high to very high, medium to high, low to medium, and very low to low flood susceptibility, have affected 7,600,500, 8,464,500, 7,766,100, and 8,257,500 m² of the watershed, respectively. On the other hand, regions with moderate

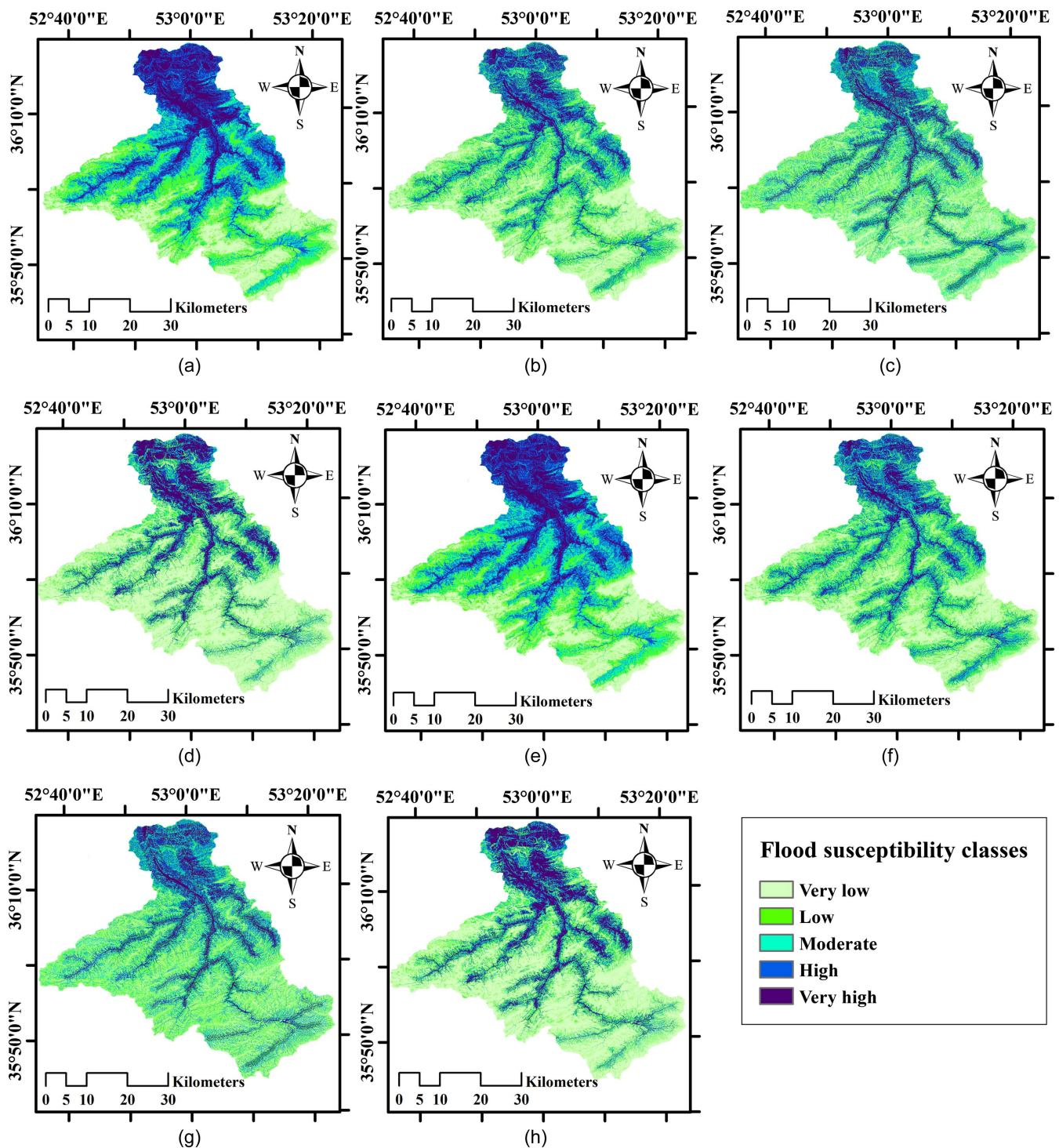


Fig. 6. Flood susceptibility map based on the algorithms of (a) ANN in 2020; (b) LR in 2020; (c) RF in 2020; (d) SVM in 2020; (e) ANN in 2050; (f) LR in 2050; (g) RF in 2050; and (h) SVM in 2050.

Table 5. The performance of ML models in producing susceptibility map

Model	RMSE	R ²
ANN	0.2834	0.8667
LR	0.1752	0.8773
RF	0.1920	0.8525
SVM	0.1252	0.9373

to very low, high to low, and very high to medium flood susceptibility, which encompass 4,500, 166,500, and 57,600 m² of the watershed, respectively, are three of the most remarkable changes to lower the level of flood susceptibility. Changes in low to very low, medium to low, high to medium, and very high to high flood susceptibility regions successively lower the level of flood susceptibility by 6,508,800, 5,638,500, 5,001,300, and 3,376,800 m² of the watershed area.

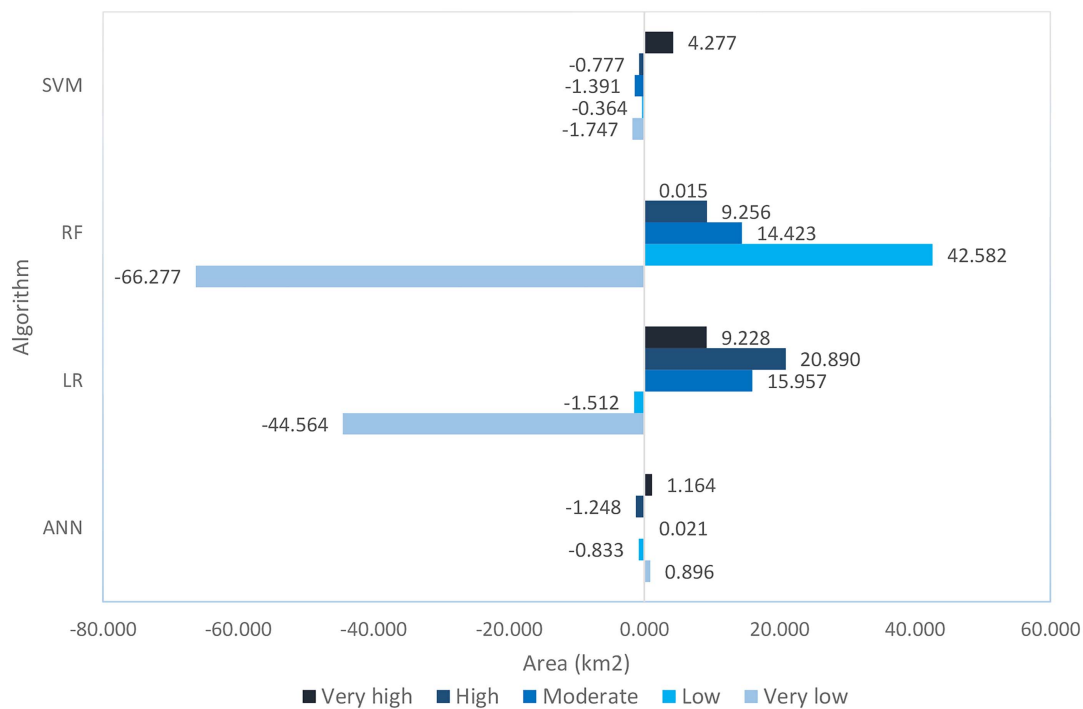


Fig. 7. The difference in the areas of each flood susceptibility class between 2020 and 2050 based on different algorithms.

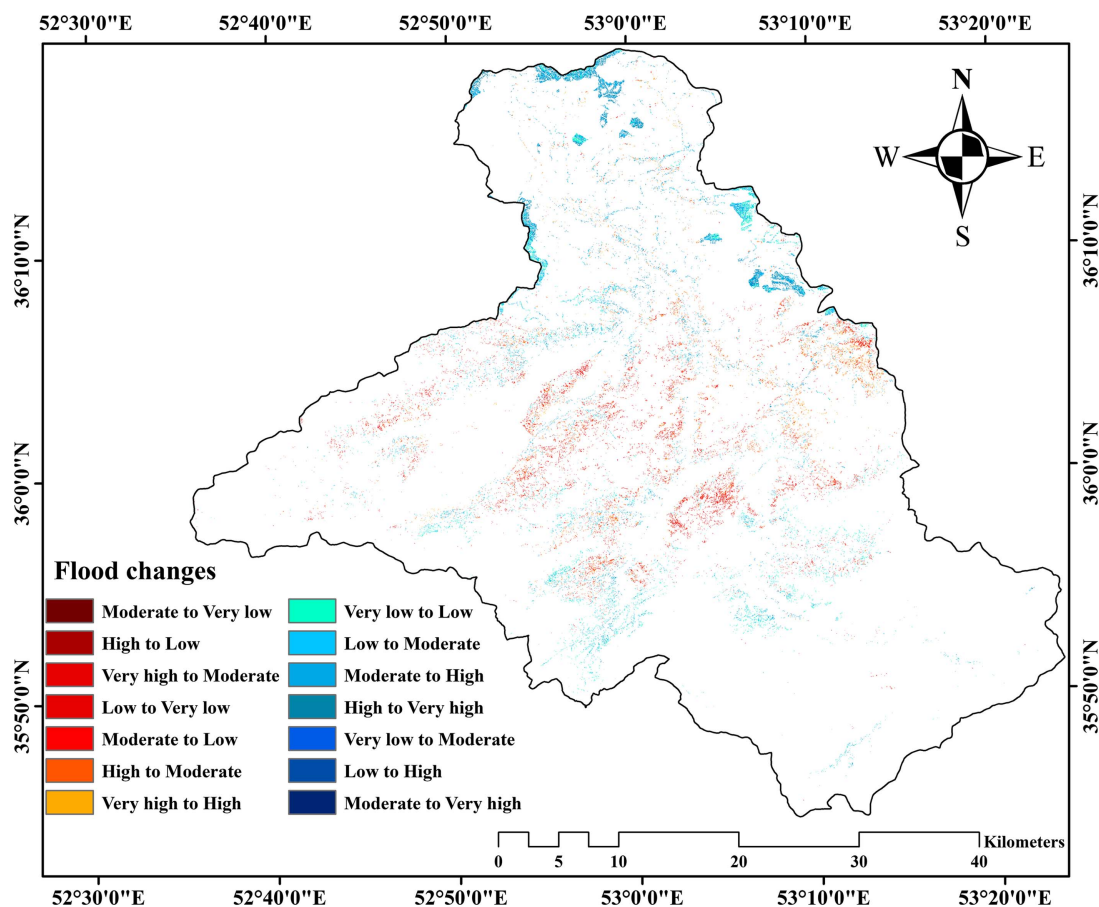


Fig. 8. Map of flood susceptibility changes from 2020 to 2050 based on SVM algorithm.

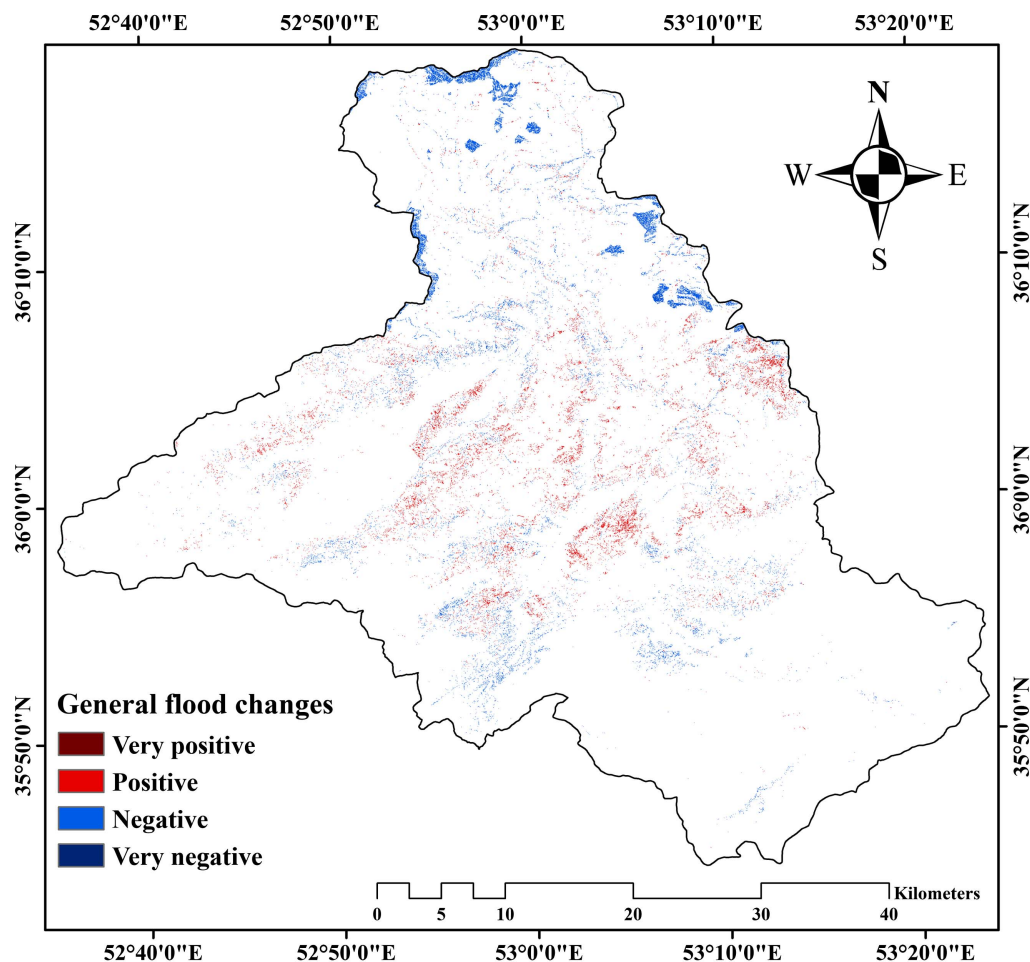


Fig. 9. Map of flood susceptibility general changes from 2020 to 2050 based on SVM algorithm.

As shown in Fig. 8, in the northern areas of the watershed, which will experience changes in LULC between 2020 and 2050, such as the conversion of forest to pasture, the level of flood susceptibility will have been increased in the future. Similarly, in the central areas, which in the LULC changes map between 2020 and 2050 have changes such as conversion of pasture to forest or conversion of bare land to pasture, the level of flood susceptibility will have been decreased by 2050.

Changes to reduce the level of flood susceptibility can be divided into two categories, including very positive and positive changes. Very positive changes are changes in which flood susceptibility is improved by at least two levels. For example, a change in the level of flood susceptibility from very high to moderate is considered as a very positive change, and changes with an improvement of one level of susceptibility are considered as positive changes. Similarly, changes in the direction of increasing the level of flood susceptibility are classified into two categories, including very negative and negative changes, where very negative changes above one level of flood susceptibility are in a destructive direction, and negative changes of a level are in the direction of increasing flood susceptibility (Fig. 9).

As it is obvious in Fig. 9, 0.229 km² of the watershed area will be subject to very positive flood susceptibility changes, 20.52 km² will be subject to positive flood susceptibility changes, 0.264 km² will be subject to very negative flood susceptibility changes and 32.089 km² of the relevant watershed area will be subject to negative flood susceptibility changes. Moreover, in general, the LULC of 20.749 km² of the watershed will change in such a way that the

level of flood susceptibility in 2050 in those areas has decreased compared to 2020 and the land cover of 32.353 km² from the watershed will have changed by 2050, which will increase the level of flood susceptibility in those areas.

Conclusion

This study aimed to investigate the effects of LULC changes on current and future flood susceptibility. In order to prepare a flood susceptibility map, four ML methods, including ANN, LR, RF, and SVM along with nine influential factors in flood, including altitude, slope, distance from rivers, LULC, TWI, SPI, profile curvature, rainfall, and soil type were used. LULC of year 2050 was predicted to forecast future flood susceptibility map and the impacts of LULC changes on flood were investigated. To predict LULC map, the LCM model including MLP algorithm and Markov chain along with ten influential factors and, the LULC maps of 2000, 2010, and real 2020 were used. The projected LULC map for 2050 shows the increase in deforestation, especially in the northern parts, and the increase in the built-up areas in the study watershed. In general, according to the prediction, the areas of forest and bare land LULCs will be declined by 15.09 and 42.12 km², respectively. Moreover, the built-up areas, agriculture land, and pasture will have increased by 11.44, 1.21, and 45.46 km², respectively. One of the most devastating changes in the flood discussion, namely the conversion of forest to pasture, would occupy a considerable area of 48.15 km² of the study watershed, according to an analysis of the LULC changes

between 2020 and 2050 and determining the conversion between various LULC classes. The northern regions of the watershed will be the major focus of these conversions. Furthermore, the change of LULC from pasture to forest with an area of 27.016 km² of the watershed in the future, will be one of the most important LULC changes in the future that will affect most of the central parts of the watershed. In general, it can be concluded that 61.008 km² of the watershed will have changes to reduce the probability of floods and 103.668 km² will have changes to increase the probability of floods. Furthermore, according to the RMSE criterion of flood susceptibility, SVM, LR, RF, and ANN algorithms showed the values of 0.1252, 0.1752, 1920.0, and 0.2834, respectively, that SVM showed the best performance among all four algorithms. Also, by R² criterion, SVM, LR, ANN, and RF algorithms with the values of 0.9373, 0.8773, 0.867, and 0.8525, respectively, all showed good performance, that similarly SVM algorithm had the highest accuracy in the prediction. The findings of all four algorithms show that the northern portions of the watershed have a larger likelihood of flood occurrence than the other regions. The prediction results of the flood susceptibility map in all four ML methods, except for the ANN method, show that in the next 30 years, the area of regions with very low flood susceptibility will decrease, and based on all algorithms, the area of regions with very high flood susceptibility will increase. The map of changes of the conversion of flood susceptibility classes from 2020 to 2050 based on the best algorithm, SVM, showed that 20.754 km² of the watershed surface will change in such a way that the flood susceptibility in those areas is reduced and 32.352 km² of the surface of the watershed will change by 2050 in such a way that the flood susceptibility in those areas will increase. In light of the fact that only LULC changes have been considered in the 2050 forecast and other influencing factors have remained constant, the negative impact of land use changes over the next 30 years is proved. Particularly the destruction of forest areas and vegetation and the increase of built-up areas and bare lands is obvious in increasing the probability of flooding and increasing areas with high and very high flood risk. In general, the results showed the efficiency of LCM method in forecasting future LULC and the efficiency of ML methods in preparing flood susceptibility map.

Data Availability Statement

All data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

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